PPP-E-540C
September 12, 1973
SUPERSEDING
Fed. Spec. PPP-E-540B
March 30, 1967

FEDERAL SPECIFICATION

ENVELOPES: WATER-RESISTANT FOR PACKING LIST AND SHIPPING DOCUMENTS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a water-resistant envelope suitable for protection of packing list and shipping documents.

1.2 Classification.

1.2.1 Classes and styles. The envelopes shall be of the following classes and styles, as specified (see 6.2).

Class 1 - Envelopes, packing list.
  Style 1 - With attachment holes and double safety fold.
  Style 2 - Without attachment holes and unfolded.
  Style 3 - With attachment holes and single safety fold closed with metal clasp.
  Style 4 - With self-adhering adhesive coated back.

Class 2 - Envelopes, shipping documents.
  Style 1 - With attachment holes and double safety fold.
  Style 2 - Without attachment holes and unfolded.
  Style 3 - Without attachment holes and single fold closed with metal clasp.
  Style 4 - With self-adhering adhesive coated back.

Class 3 - Envelopes, important papers.
  Style 1 - With attachment holes and double safety fold.

Class 4 - Envelopes, material, release/receipt documents.
  Style 1 - With attachment holes and double safety fold.
  Style 2 - Without attachment holes and unfolded.
  Style 3 - With attachment holes and single safety fold closed with metal clasp.
  Style 4 - With self-adhering adhesive coated back.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitations for bids form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
UU-P-268 - Paper, Kraft, Untreated, Wrapping.
UU-P-270 - Paper, Wrapping, Waxed (Dry).
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).
Fed. Test Method Std. No. 101 - Preservation, Packaging, and Packing Materials; Test Procedure.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

FSC 8105

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Envelopes, classes 1, 2, 3 and 4; styles 1, 2 and 3. Unless otherwise specified the envelopes shall be fabricated from one of the following water-resistant materials:

(a) Waterproofed kraft wrapping paper conforming to PPP-B-1055 class C-1, basis weight 30-50-30.

(b) Dry waxed wrapping paper, basis weight 50 to 60 pounds (24 x 36-500 sheets) conforming to UU-P-270, type II, class 3.

(c) A plastic film (polyethylene thin gage) conforming to L-P-378 laminated between two sheets of kraft untreated paper conforming to UU-P-268, grade B, 30 pounds basis weight (24 x 36-500 sheets).

(d) A heat sealable barrier material comprising a minimum of 15 pounds of low-density polyethylene extrusion coated to natural kraft paper conforming to UU-P-268, grade B, 50 pounds basis weight (24 x 36-500).

3.1.2 Envelopes, classes 1, 2 and 4; style 4. The envelopes shall be fabricated from transparent polyethylene film (3.1.2.1), 0.003 inch thick, conforming to L-P-378, except that the impact resistance requirement in table II thereof shall be 45 percent of the values shown. The film in combination with the adhesive shall be 0.004 inch thick minus 10 percent to 0.005 inch thick plus 10 percent. The envelope shall be coated with a self-adhering adhesive (3.1.2.2).

3.1.2.1 Color. Envelopes delivered under any contract resulting from this invitation must be a good commercial grade to match the Government sample with respect to color only.

3.1.2.2 Adhesive. The polyethylene film shall be coated on oneside with a pressure sensitive adhesive requiring no moisture, heat or other special preparation prior to its application. The adhesive shall be protected with a carrier sheet (interliner), conforming to requirements of 3.1.2.4, and shall be capable of being removed merely by stripping without exhibiting edge tear, paper delamination, or adhesive delamination, lifting or parting.

3.1.2.3 Transparency. The film in its nonadhesive coated areas, shall be sufficiently clear and transparent to allow easy reading of 10 point black type when it is applied directly over it.

3.1.2.4 Protective interliner (carrier sheet). The carrier sheet shall adhere uniformly and equally to the adhesive coated areas of the polyethylene film and shall be securely attached. It shall conform to the adhesion requirements of table I, as tested in accordance with 4.4.3. The carrier sheet shall be a specially treated kraft stock suitable for this purpose of 80 pound (24 x 36-500) basis weight plus or minus 10 percent.

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3.2 Sealing material-envelopes, classes 1, 2, 3 and 4, styles 1, 2, 3 and 4. The sealing material shall be such that, in the dry condition and after water spray cycle when tested as specified in 4.4.1, the sheets sealed together cannot be separated manually without damage to the surface of the paper united to form the seal.

3.2.1 Envelopes made from material specified 3.1.1. The envelopes shall be formed, using two sheets or folding one sheet, by placing heat sealable surfaces together and heat sealing seams along two sides and bottom edge.

3.3 Water-resistant.

3.3.1 Envelopes, styles 1, 2 and 3. The envelope shall protect the packing list from water penetration, when tested as specified in 4.4.1. At expiration of the spray cycle specified, the condition of the envelope shall not show evidence of water penetration.

3.3.2 Envelopes, style 4. The envelope shall protect the packing list from water penetration, when tested as specified in 4.4.1. At expiration of the spray cycle specified, the condition of the envelope shall not show evidence of water penetration.

3.4 Construction.

3.4.1 Envelopes, style 1. The envelope shall be open end with a seal 1 ± 1/8 inch wide along each of the two long sides. One end shall be closed and sealed with a seal 3/4 ± 1/8 inch wide. There shall be no other seams.

3.4.1.1 Closures. The envelope shall be lightly scored 7/8 ± 1/8 inch from, and parallel to, the open end of the envelope to provide the first fold. Another scoring shall be made 1 ± 1/32 inch below and parallel to the first scoring to provide for a second fold. In closing the envelope, the double safety fold shall be in a direction away from the face or printed side of the envelope.

3.4.1.2 Holes. Six holes shall be centered along the sealed border of the envelopes, three on each side, extending down from the end flap. The holes shall be located so that when the double safety fold is closed on the scored lines, each of the two sets of three holes coincide to form in effect, two single holes in the closed envelope, one at either end of the double safety fold. The diameter of each of the six holes shall be not less than 1/8 or greater than 3/16 inch.

3.4.2 Envelopes, style 2. The envelope shall be open end with a center overlap seam on the long side opposite the face. The front, or face, shall be the side opposite the overlap seam. One end shall be closed and sealed with a seam 3/4 ± 1/8 inch with flap type closure.

3.4.3 Envelopes, style 3. The envelope shall be open end with a center overlap seam of 5/8 ± 1/8 inch on the long side opposite the face. The front, or face, shall be the side opposite the overlap seam. One end shall be closed and sealed with a seam 3/4 ± 1/16 inch wide. The open end shall be provided with a flap type closure by extending the long side opposite to the center seam 1 1/2 ± 1/8 inch beyond the body of the envelope. Alternatively, envelopes may be open end with heat sealed side seams and bottom seam not less than 1/2 inch ± 1/8 inch wide.

3.4.3.1 Closure. The envelope shall be lightly scored 2 ± 1/32 inch from, and parallel to, the end of the flap. The flap shall be provided with 5/16 inch tab reinforced hole, centered 5/8 ± 1/16 inch from end of flap. A metal clasp shall be secured in place on the center seam side of the envelope in a location so that when the envelope is folded at the score line the clasp and hole in the flap will coincide.

3.4.3.2 Holes. Four holes shall be located in the corners of the envelopes within the border of the scored line and sealed end. The holes shall be centered 3/8 ± 1/8 inch from the sides. Two of the holes shall be centered 3/4 ± 1/8 inch from the scored line, the other two holes shall be centered 1/2 ± 1/8 inch from the end. Each hole shall have a diameter of 3/16 inch, with a plus tolerance not greater than 1/16 inch. The four holes shall be provided with metal eyelets 3/16 ± 1/32 inch inside diameter.

3.4.4 Envelopes, style 4. The envelope shall be formed from a single flat sheet, folded with side seams heat sealed parallel with each other and with the outer edge of the envelope, or by two flat sheets heat sealed parallel sides and bottom. The back shall be coated with a pressure sensitive, self-adhering adhesive, protected by a treated paper backing of the same size.

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3.4.4.1 Holding power. When tested in accordance with 4.4.2, the product shall conform to the requirements of table I and shall not exhibit adhesive transfer in excess of 20 percent by weight to the test panel.

3.4.4.2 Adhesion. When tested in accordance with 4.4.3, the product shall conform to the requirements of table I.

Table I.

I--Adhesion-180° peel back stainless steel (oz. or g., as specified per/1 inch width). (Oz./inch)	Minimum	Maximum
Initial to S/A	25 oz.	--
*Initial to carrier	15 g.	60 g.
Initial to backing	14 oz.	--
After accelerated aging:		
To S/S	25 oz.	--
*To carrier	15 g.	60 g.
To backing	14 oz.	--
*Withdrawal speed of jaws-50 inch minute.		
II-Holding power-90° peel.	Minimum	Maximum
Initial to glass after accelerated	600 sec.	---
Aging to glass (in sec. to travel 4 inches with 1000 g. load	600 sec.	---

3.4.4.3 Unless otherwise specified, the flap shall extend a minimum of 3/4 inch beyond the opening edge and shall have a self-adhering adhesive 1 1/4 + 1/8 inch wide uniformly and continuously applied to the outer surface of the flap. The adhesive shall be protected by a strip of treated paper 1 1/2 inches wide and full width of the envelope. As an alternative, the flap may be on the outer (nonadhesive) side of the envelope and extend beyond the inner side, minimum of 1/4 and maximum of 3/8 inch. The outer side of the envelope shall have a dotted line and the words "Fold Here" 1/2 inch from the outer edge of the flap.

3.5 Dimensions and sizes.

3.5.1 Styles 1, 2, 3 and 4. Envelope dimensions specified shall apply to the overall clear inside length and width, exclusive of flaps, seals, or attachment holes. For purpose of permitting the maintence of stocks of envelopes and for references, common sizes are listed in table II. Envelopes; however, shall be supplied in dimensions specified in the contract or order.

Table II.

Size designation	Inside dimensions in inches[1]
1	5 by 3 3/4
2	5 3/4 by 4 1/4
3	6 1/2 by 4 1/2
4	7 by 4 1/2
5	8 1/2 by 5
6	8 1/2 by 6 1/4
7	8 3/4 by 6
8	9 1/4 by 6 3/4
9	12 by 10

[1]Dimensions are expressed in inches and fractions thereof with length first and width open end second as follows: 6 1/2 by 4 with a tolerance plus or minus 1/8 inch.

3.5.2 Style 4. Envelope dimensions specified shall apply to the inside length and width, exclusive of flaps, seals, or seams. For the purpose of permitting the maintenance of stocks of envelopes and for reference common sizes are listed in table III. Envelopes, however, may be supplied in other dimensions as specified in the contract or order.

Table III.

Size designation	Description	Inside dimensions in inches[1]
10	Controlled data card	8 1/2 by 3 3/4
20	GSA form 308	7 by 4 3/4
30	GSA DD 250 size	9 1/8 by 8
40	Material release/receipt document	8 1/2 by 6 1/4

[1]Tolerance on sizes plus or minus 1/8 inch.

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3.6 Printing. Unless otherwise specified (see 6.2), a legend shall be printed midpoint of, parallel to the long sides and centered on the face of the envelope.

3.6.1 Class 1, envelopes, packing list. Unless otherwise specified by the contracting officer (see 6.2), the legend shall be as follows:

PACKING
LIST
ENCLOSED

When specified, the legend shall be printed in one line parallel to the long sides, 1-1/2 inches from the top side as follows:

PACKING LIST ENCLOSED

3.6.2 Class 2, envelopes, shipping documents. Unless otherwise specified by the contracting officer, the legend shall be as follows:

SHIPPING
DOCUMENTS
ENCLOSED

3.6.3 Legend. The specified legend shall appear a minimum of once on each envelope, and shall be clear, legible, in waterproof plain block letters 1/2 or 3/4 ± 1/8 inch high, as specified by the procuring activity.

3.6.4 Class 3, envelope, important papers. The legend shall be as follows:

NOTICE
THIS ENVELOPE CONTAINS
IMPORTANT PAPERS
DO NOT DESTROY

3.6.4.1 Legend. The legend shall appear centered in each of the two halves of the face of the envelope, the center to be created by an imaginary line connecting the midpoints of the long sides of the envelopes and shall be clear, legible and waterproof, plain black block letters, 1/4 inch to 3/8 inch high. The printing shall appear once on all envelopes 4 by 5 inches or smaller, and twice on all envelopes larger than 4 by 5 inches.

3.6.4.2 The face of the envelopes shall be printed along the borders, excluding the open end, in solid or shaded black to a width of 7/8 inch. When a shaded area is provided, there shall be at least 8 lines per inch running diagonally through the width of the border. The legend "fasten through shaded area only", shall appear adjacent to the border along each side and the closed end, in letters approximately 1/8 inch high.

3.6.5 Class 4, envelopes, material, release/receipt documents. The envelope dimensions shall be as specified in table II. The legend shall be as specified in 3.6.3 and as follows:

MATERIAL RELEASE/RECEIPT

DOCUMENTS ENCLOSED

The legend shall be printed parallel to the sides of the envelope, on the face of the envelope, and not less than 1/4 inch nor more than 1/2 inch from the edge of the long sides. The center area on the face of the envelope shall not be obstructed.

3.7 Workmanship. The envelope shall be uniformly constructed free from pinholes, tears, cuts, sharp creases, or wrinkles. The finished product shall conform to the levels of quality established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same class and style, made by the same process from the same compound material by one manufacturer, and submitted for delivery at one time.

4.2 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. An examination of the end item shall be made in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQL's) set forth below. A random sample shall be drawn from each lot of material for each class and style offered for acceptance for visual and dimensional characteristics. For purposes of determining the sample size in accordance with MIL-STD-105, the lot size for examinations under 4.3.1.1 through 4.3.1.3, shall be expressed in units of packages of envelopes of the specified class, style, of the same material, and for examinations under 4.3.1.4, shall be expressed in units of shipping containers.

4.3.1.1 Examination of the end item for defects in appearance, construction, and general workmanship. The sample unit for this examination shall be one envelope. Examine both sides.

Examine	Defects
Appearance and workmanship	Tear, cut or holes (including pinholes). Presence of scuff mark, chafed spot, sharp wrinkles, or creases in body or sealed areas.
	Envelopes stick together to the extent that opening causes tearing or injury to any surface.
	Seams not uniform and in alignment; not completely sealed throughout their entire area. Seams other than specified.
Construction Style 1	Not of uniform construction; open end along one short side of envelope.
	One end not equipped with flap and double safety fold.
	Folds not lightly scored with 2 score line parallel to each other and end of flap.
	Double safety fold, in closing of envelope, not made in direction away from face or printed matter.
	Does not consist of two side seams (1 on each side) and one bottom or end seam.
	Not provided with 6 attachment holes consisting of 3 holes on each side centered along seam border.
	Attachment holes not centered along the sealed area in such manner that when the double safety fold is closed on scored lines, each of the two sets of 3 holes do not coincide to form, in effect 2 single holes in the closed envelope, 1 at each end of the double safety fold.
	Attachment holes not provided.
Style 2	One end not equipped with unfolded flap. Does not consist of one sealed overlap seam on back and one bottom or end seam.
	Presence of any attachment hole(s).
Style 3	One end not equipped with flap and single safety fold.
	Does not consist of one overlap seam on back and one bottom or end seam.
	Fold not lightly scored with single score line parallel to end of flap.
	Safety fold, in closing of envelope, not made in direction away from face or printed matter.
	Flap not provided with a centered tab reinforced hole.
Backle	Back not provided with a securely attached metal clasp centered in overlap seam area.
	Metal clasp not centered in seam area in such a manner that when envelope is folded on score line clasp and hole in flap do not coincide.

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(continued)

Examine	Defects
	Not provided with 4 attachment holes consisting of 2 on each side positioned between score line and sealed end.
	Attachment holes not provided with metal eyelets.
Style 4	Self-adhering adhesive not uniformly applied.
	Back of envelope not provided with a securely attached paper backing.
	Color not as specified.
Printing	Not printed in manner specified for the applicable class envelope; omitted, illegible, incorrect, or not in position specified (see 3.6).
	Legend lettering not block type; not of uniform size.

4.3.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one envelope.

Examine	Defects
Inside clear dimensions	Length or width, exclusive of flap, vary by more than plus or minus 1/8 inch from dimensions specified.
	Diameter of tab reinforced hole in flap greater than 11/32 or less than 9/32 inch.
	Top or bottom attachment holes not centered in position specified within allowable tolerance.
	Diameter or attachment holes not within allowable tolerance.
Style 4	Size of protective backing not of dimensions specified.
	Flap greater than 1 3/8 or less than 1 1/8 inch, when specified as required in 3.4.4.3.
Legend	Height of lettering varies by more than plus or minus 1/8 inch from size specified (see 3.6.3).

4.3.1.3 Examination of count per unit container, as applicable. The sample unit for this examination shall be one package or unit container of envelopes. The average count per package or unit container shall be not less than the specified or indicated quantity.

4.3.1.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and markings comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined fo closure defects.

Examine	Defects
Packaging	Not level specified; not in accordance with contract requirements.
	Not packaged in units of multiples specified for applicable size envelope.
	Unit packages not packaged as specified.
	Mixed classes, styles, or sizes in same unit package.
	Unit containers, when required, not snugly packed; contain fillers or waste space.
	Packaging material not as specified; closures not accomplished by specified or required methods or materials.
Packing	Not level specified; not in accordance with contract requirements.
	Unit packages or containers not snugly packed.
	Arrangement or number of unit packages or containers per shipping container not in accordance with contract requirements.
	Container material not as specified; closures not accomplished by specified or required methods or materials.
Strapping (steel or tape)	Not properly placed lengthwise, girthwise, or horizontally; not applied in manner specified.
	Tension too great, i.e., strapping tears or cuts through facing of container.
Marking	Interior or exterior markings (as applicable) illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements.
Weight	Gross weight exceeds contract requirements for the shipping container.

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4.3.1.5 Inspection levels and AQL's for examinations. The inspection levels for determining the sample size and the AQL's expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQL
4.3.1.1	I	2.5
4.3.1.2	S-3	2.5
4.3.1.3	S-2	-
4.3.1.4	S-2	2.5

4.3.2 Testing of end item. The end item shall be tested for the applicable characteristics, as indicated in table IV, for each lot presented for examinations for each class and style, of the same material. The sample unit for test shall be 2 envelopes. Each sample unit for test shall be drawn from a different package, or unit container, and insofar as practicable shall be taken from a different shipping container for a better representation of the lot. Five sample units, randomly selected, shall be tested with no evidence of failure to meet the specified requirements.

4.3.2.1 Polyethylene laminated sheet. This requirement, as specified in 3.1.1(c), will be by acceptance of the supplier's actual physical testing or by certified test data.

4.3.2.2 Polyethylene coated kraft. This requirement, as specified in 3.1.1(d), will be by acceptance of the supplier's actual physical testing or by certified ~~test~~ data.

Table IV. Instructions for testing (sample units)

Characteristic	Specification reference Rqmt. par.	Specification reference Test method	Requirements applicable to individual units	Number determinations per sample unit	Results reported as1
Seam strength:					
Dry condition	3.2	4.4.1	X	1	Pass or fail
Wet condition	3.2	4.4.1	X	1	Pass or fail
Water resistance	3.2	4.4.1	X	1	Pass or fail

1If failure is indicated, report description of failure.

4.4 Test methods.

4.4.1 Water resistance, envelope styles 1, 2, 3 and 4. Before starting the water resistance test, insert a packing list or a reasonable facsimile thereof, in the envelope. Style 1 envelope shall be double safety fold closed and then sealed at each of the two attachment holes with cellophane or masking tape extending a maximum 1 inch around each edge to prevent opening. Styles 2 and 3 envelopes, after closing, shall be sealed along the flap edge by means of application of a strip of waterproof tape. Style 4 envelope, remove carrier sheets from flap, and from back of envelope. Fold envelope back 180° so that flap area is sealed and the envelope is adhered to a test panel 12 by 12 inches conforming to the requirements of method 2050 of Fed. Test Method Std. 101, for surface finish. In an open tap reservoir, place the closed envelope flat on a false raised floor of wooden slats, below a spray type nozzle located 2 feet above the envelope. The design of the spray nozzle shall be such as to break up the flow of water into a heavy mist continuously at a water temperature of 75° ± 5° F., for a period of 4 hours. Sufficent water pressure shall be applied so the 2.5 ± 0.5 gallon per hour are uniformly distributed over a floor area of 1 square foot. The use of fresh tapwater instead of a recirculation system is permissible as long as the spray control and specified temperature are achieved. Upon completion of the water spray period, the envelope shall be examined for conformance to 3.2 and 3.3.

4.4.2 Holding power, envelope style 4. A 4 by 6-inch plate glass panel, minimum 1/4 inch thick is cleaned according to method 2050. A 2 by 6 inch adhesive coated sample with backing (A) is accurately weighed on an analytical balance to the nearest tenth of a milligram. The backing (B) is then removed and weighed. The adhesive coated sample is applied, adhesive face against the glass, as described in method 2050 of Fed. Test Method Std. 101. A 1-inch section is lifted and overlapped 1/2 inch face to face. A 1000g load is attached and the time in seconds is measured for the 2 inch strip to travel 4 inches (see diagram I). At the end of the 4 inch travel, the 4-inch portion of the release film (D) is cut and accurately weighed. The amount of adhesive transferred is calculated as described below.

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(A-B) = C (Film and adhesive on 2 inch by 6 inch sample before test).

(2/3 C) - Film and adhesive on the 2 inch by 4 inch area before test.

(2/3 C) = D = Amount of adhesive transferred.

The total weight of adhesive (E) is obtained by stripping a separate 2-inch by 4-inch piece (backing removed) by means of a suitable solvent such as acetone using a soxhlet or by other means.

The weight of the undissolved material is subtracted from the weight of 2/3C to determine the total weight of adhesive.

The percent of transferred adhesive is calculated as follows:

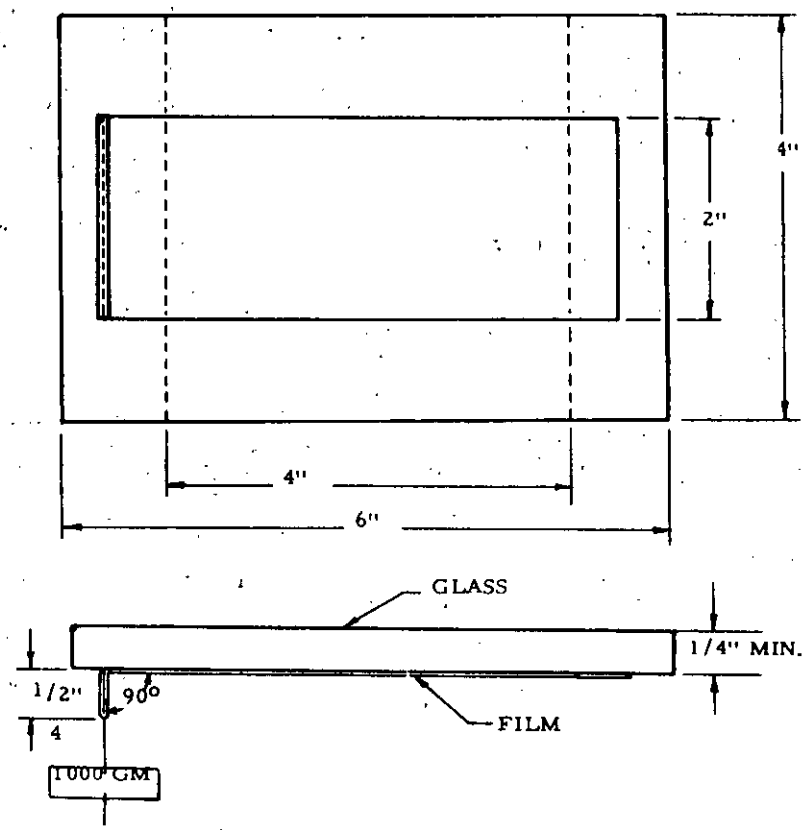
$$\frac{(2/3C - D}{E} \times 100 = \%$$

4.4.3 Adhesion. The following test shall be conducted as described in Fed. Test Method Std. No. 101. Submissions for test shall include twelve, 8 by 10 inch sheets of flat coated stock printed prior to envelope formation.

Test	Method	Exceptions
Adhesion	2050	None
Adhesion to carrier sheet	2050	Substitute carrier sheet for stainless steel panel.
Adhesion to backing (polyethylene sheet)	2050	Specimen shall be applied to film backing surface of film covered steel panel.

Conformance shall be both to initial and after accelerated aging of 14 days at 140°F. (see table I).

DIAGRAM I



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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the envelope of the same class, style, and material shall be unit packaged in increments of 50 or 100, as specified (see 6.2). The package shall be made in one of the following ways: by placing the envelopes in a carton or box; by wrapping the envelopes securely in kraft paper; or by banding securely. Unless otherwise specified, envelopes shall be packaged flat without folding. When carton or boxes are used, there shall be no fillers or waste space. Wrapped or banded packages shall be secured with tape or adhesive.

5.1.2 Level B. One hundred envelopes of the same class, style and material shall be packaged by placing the envelopes in a snug-fitting carton or box, by wrapping securely in kraft paper, by banding or a polyethylene bag. The packaged envelope shall be secured within the container, wrap or band to prevent accidental opening or spilling during shipment and storage.

5.1.3 Level C. The envelopes shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Envelopes of one class and style material only, packaged as specified in 5.1, shall be packed in a snug-fitting box conforming to PPP-B-636, class weather-resistant, grade V2s. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of the container specification. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B. Ten unit packages (1000 envelopes) of one class, style and material shall be packed in a box conforming to PPP-B-636, type, class and style optional. Bursting strength of the fiberboard shall be not less than 275 p.s.i. Each box shall be closed in accordance with the appendix to the box specification.

5.2.2.1 When specified (see 6.2), the shipping container shall be a Grade V3c, V3s or V4s box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification.

5.2.3 Level C. Envelopes shall be packed in a manner that will afford adequate protection against damage. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation.

5.2.4 Standard pack (civil agency procurement). The standard pack for civil agency procurement shall be as specified in 5.1.2 and 5.2.2 (see 6.2).

5.3 Marking.

5.3.1 Civil agencies requirements. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies requirements. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use, envelope styles 1, 2, 3 and 4. The envelopes covered by this specification are intended as a protective medium for shipping documents and packing lists.

(a) Styles 1, 2 and 3 are to be tacked, stapled, tied, or secured by application of a protector to the exterior of the shipping containers.

(b) Style 4 is to be applied directly, without the application of staples, tacks, ties or any other means except that it shall be self-adhering.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Class, style and material, as specified (see 3.1.1).
(c) Quantity required.
(d) Size or envelope, inside dimensions (see 3.5).
(e) Color other than specified (see 3.1.2.1).
(f) Legend, location, position and size of lettering, if other than specified (see 3.6).
(g) Levels of packaging and packing required (see 5.1 and 5.2).

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 69

Preparing Activity

GSA-FSS

Military Coordinating Activities:

Army - GL, MU, MI
Air Force - 69, 82
Navy - WP

User Activities:

Navy - SH, MC, Y
Army - MO
Air Force - 71

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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